

Quick review of 1-variable derivative.

Consider $f: \mathbb{R} \rightarrow \mathbb{R}$.

The derivative $f'(a)$, of f at a , is defined by

$$f'(a) = \lim_{x \rightarrow a}$$

$$\frac{f(x) - f(a)}{x - a}$$

abs. change in f .

abs. change in argument.

relative change
in value of f .

$$= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Let's see this now in a multi-variable setting. Consider a function f of two variables. $f: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$
 $(x, y) \mapsto f(x, y)$

There are two partial derivatives for f . $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$

same second
argument

for partials
use ∂
instead of d

$$\left. \frac{\partial f}{\partial x} \right|_{(a,b)} = \lim_{x \rightarrow a} \frac{f(x, \underline{\underline{b}}) - f(\underline{\underline{a}}, b)}{x - a}$$

$$\text{OR } \lim_{h \rightarrow 0} \frac{f(a+h, b) - f(a, b)}{h}$$

and similarly

$$\left. \frac{\partial f}{\partial y} \right|_{(a,b)} = \lim_{h \rightarrow 0} \frac{f(\underline{\underline{a}}, b+h) - f(\underline{\underline{a}}, \underline{\underline{b}})}{h}$$

fixed a
↓

First example

Consider the function defined by

$$f(x, y) = x^2 y$$

$$\frac{\partial f(a,b)}{\partial x} = \lim_{h \rightarrow 0} \frac{f(a+h, b) - f(a, b)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(a+h)^2 b - a^2 b}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{a^2 b} + 2a h b + h^2 b - \cancel{a^2 b}}{h}$$

$$= \lim_{h \rightarrow 0} (2ab + hb)$$

$$= 2ab, \quad \text{using linearity of limits.}$$

linearity

AND

$$\frac{\partial f}{\partial y} = \lim_{h \rightarrow 0} \frac{f(a, b+h) - f(a, b)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{a^2 (b+h) - a^2 b}{h}$$

$$= \lim_{h \rightarrow 0} \frac{a^2 b + a^2 h - \underline{a^2 b}}{h}$$

$$= \lim_{h \rightarrow 0} a^2$$

$$f(x, y) = x^2 y$$

$$= a^2$$

True for all a, b so we may
as well ~~write~~ write

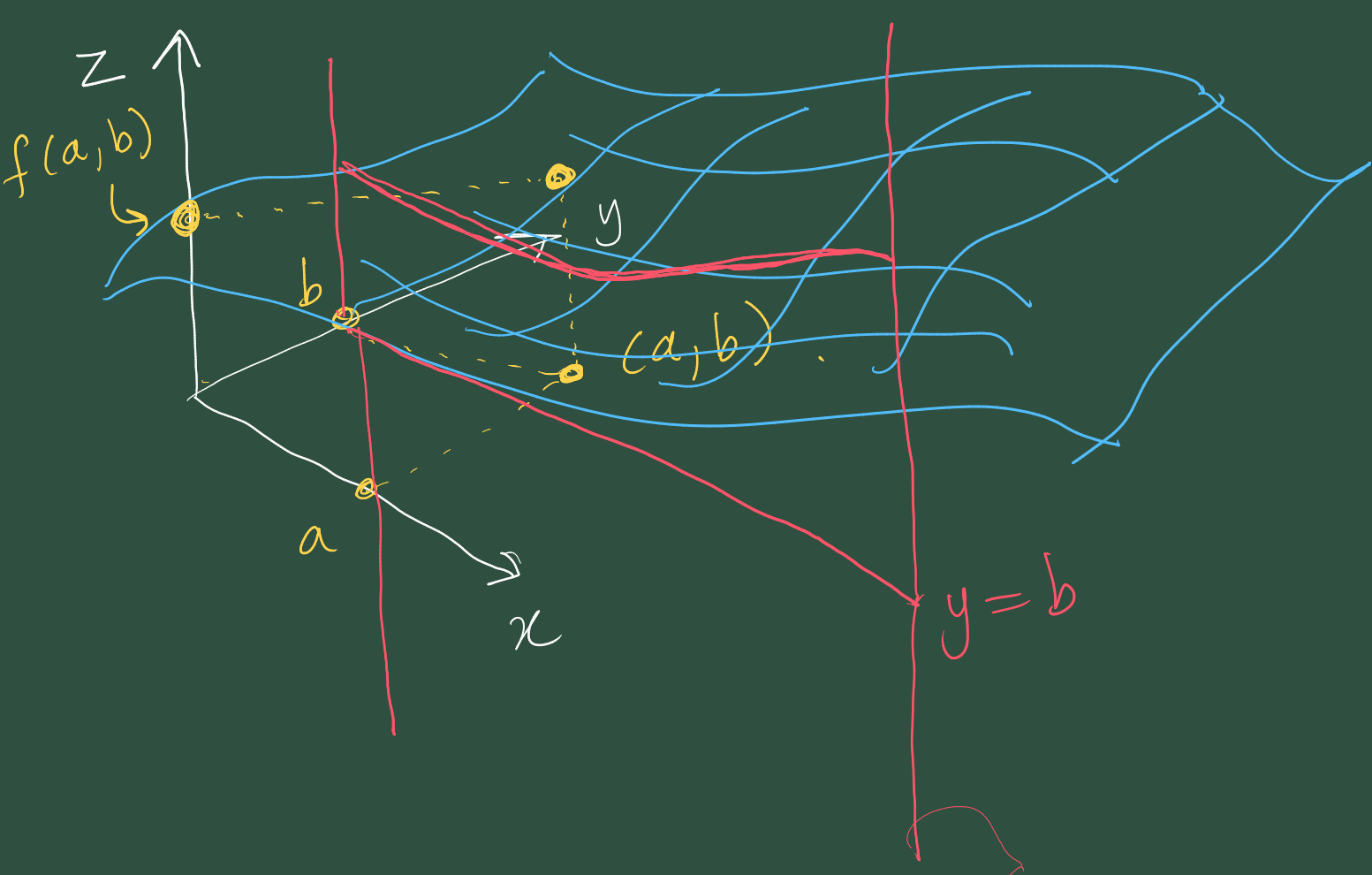
$$\frac{\partial f}{\partial x} = 2xy, \quad \frac{\partial f}{\partial y} = x^2$$

These expressions seem familiar
from the ~~derivat~~ derivative process
we know from 1-variable.

Consider 2.2.3. Graphical
interpretation in textbook.



$$z = f(x, y)$$



Important principle

To differentiate a function f with respect to one variable x say, treat the other variables as fixed constants and go ahead and differentiate with respect to x using all the familiar 1-variable results and techniques.

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Example

Consider $f(x, y) = x^2 y^3 \tan(2x)$

$$\frac{\partial f}{\partial x} = y^3 \left(2x \tan(2x) + x^2 \sec^2(2x) 2 \right)$$

using $\frac{d}{dz} \tan(z) = \sec^2(z)$

$$= \frac{1}{\cos^2(z)}$$

and product and chain rules,

$$= 2y^3 (x \tan(2x) + x^2 \sec^2(2x))$$

$$\frac{\partial f}{\partial y} = 3y^2 x^2 \tan(2x)$$

Higher order partial derivatives.

We can differentiate derivatives to get higher order derivatives.

For a function $f(x, y)$ there are four possible second-order partial derivatives.

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right)$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right)$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)$$

Clairaut's
Theorem

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}.$$

and generalizes to more variables.

if we had $g(x, y, z)$

then for instance

$$\frac{\partial^2 g}{\partial x \partial z} = \frac{\partial^2 g}{\partial z \partial x} \quad \text{and so on.}$$

and to higher-order.

$$\frac{\partial^3 g}{\partial n^2 \partial z} = \frac{\partial}{\partial n} \left(\frac{\partial}{\partial n} \left(\frac{\partial g}{\partial z} \right) \right) \quad (1)$$

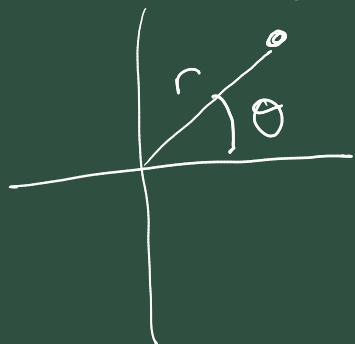
$$\frac{\partial^3 g}{\partial n \partial z \partial n} = \frac{\partial}{\partial n} \left(\frac{\partial}{\partial z} \left(\frac{\partial g}{\partial n} \right) \right) \quad (2)$$

(1) = (2) By Clairaut's theorem.

CLP3 Sec 2.3

Q6. $f(r, \theta) = r^m \cos(m\theta)$

(r, θ) using polar coordinates.



a). First find $\frac{\partial f}{\partial r}$, $\frac{\partial f}{\partial \theta}$

$$\frac{\partial f}{\partial r} = m r^{m-1} \cos(m\theta)$$

$$\frac{\partial f}{\partial \theta} = -m r^m \sin(m\theta)$$

Now second order.

$$f_{rr} = \frac{\partial^2 f}{\partial r^2} = \frac{\partial}{\partial r} \left(\frac{\partial f}{\partial r} \right)$$

$$= \frac{\partial}{\partial r} (m r^{m-1} \cos(m\theta))$$

$$= (m-1) m r^{m-2} \cos(m\theta)$$

, provided $m \geq 2$,

whereas if $m=1$

$$f(r, \theta) = r \cos(\theta)$$

$$\frac{\partial f}{\partial r} = \cos(\theta), \quad \frac{\partial^2 f}{\partial r^2} = 0$$

$$f_{\theta\theta} = \frac{\partial^2 f}{\partial \theta^2} = \frac{\partial}{\partial \theta} \left(\frac{\partial f}{\partial \theta} \right)$$

$$= \frac{\partial}{\partial \theta} (-m r^m \sin(m\theta))$$

$$= -m^2 r^m \cos(m\theta)$$

$$\left[f_{r\theta} = \frac{\partial^2 f}{\partial r \partial \theta} = \frac{\partial}{\partial r} \left(\frac{\partial f}{\partial \theta} \right) \right]$$

$$= \frac{\partial}{\partial r} \left(-m r^m \sin(m\theta) \right)$$

$$= -m^2 r^{m-1} \sin(m\theta)$$

(b) Assume $m \geq 2$.

Let's test $f(r, \theta) = r^m \cos(m\theta)$ in the PDE and see when it is a solution.

$$(m-1)m r^{m-2} \cos(m\theta)$$

$$+ \frac{\lambda}{r} m r^{m-1} \cos(m\theta) + \frac{1}{r^2} (-m^2 r^m \cos(m\theta))$$

$$= 0$$

$$\Leftrightarrow \cos(m\theta) \left((m-1)m r^{m-2} + \lambda m r^{m-2} - m^2 r^{m-2} \right)$$

$$\Leftrightarrow r^{m-2} \cos(m\theta) \left((\lambda-1)m \right) = 0$$

Notice that LHS is always zero,
only when $\lambda = 1$.